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Different facets of diversity reveal different patterns of fish assemblages in tropical rivers: The Buyuan River (China)

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ABSTRACT

Exploring the patterns of biodiversity across multiple facets of river ecosystems and their driving factors is crucial for biodiversity conservation. As the most important tributary of the lower Lancang River, the Buyuan River remains largely unexplored in terms of systematic research on its fish diversity patterns and driving factors. Here, we explored the fish taxonomic, phylogenetic, and functional diversity at both alpha and beta levels and their driving factors in the Buyuan River basin. Results showed that fish alpha and beta diversity across different facets in the Buyuan River basin exhibit distinct patterns, and the relationships among various facets of diversity had a degree of independence. Univariate Spearman correlation analysis with FDR correction showed that environmental variables and spatial variables jointly shaped the alpha diversity patterns of fish across three facets in the Buyuan River. Redundancy analysis and variation partitioning analysis indicated that the combined effects of spatial and environmental factors shaped the taxonomic and functional beta diversity patterns of fish in the Buyuan River. Specifically, environmental factors explained more of the variation in taxonomic beta diversity (adjusted R^2 = 0.364) and its turnover components (adjusted R^2 = 0.338) than did spatial factors (all adjusted R^2 values were close to 0), whereas environmental factors (adjusted R^2 = 0) explained less of the variation in the nestedness components of taxonomic beta diversity than did spatial factors (adjusted R^2 = 0.042). Similarly, functional beta diversity exhibited the same pattern. However, phylogenetic beta diversity (adjusted R^2 = 0) and its turnover (adjusted R^2 = 0.056) and nestedness (adjusted R^2 = 0) components were insensitive to both environmental and spatial factors.

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This study emphasizes the necessity of employing multifaceted diversity indicators in biodiversity assessments, as different facets of diversity provide complementary information.

1. Introduction

The current sixth global biodiversity crisis is driving unprecedented changes in freshwater ecosystems, which are essential for human sustenance (Zhou et al., 2025). Under the combined effects of global climate change and human activities, freshwater biodiversity is declining at an alarming rate. Notably, freshwater biodiversity is diminishing at a more acute rate than terrestrial and marine biodiversity (Cardinale et al., 2012; Collen et al., 2014). To successfully mitigate the reduction in freshwater biodiversity, it is essential to comprehend the impact of various environmental stressors on the composition of river biota for the purposes of biodiversity and environmental conservation (Liu et al., 2023).

Historically, numerous extensive studies on freshwater biodiversity solely account for taxonomic diversity (TD) (Liu et al., 2023). However, TD overlooks the ecological differences (evolutionary relatedness and functional differences) among species, treating all species as equivalent and independent (Jarzyna and Jetz, 2016; Pavoine and Bonsall, 2011). Assessing biodiversity exclusively through TD results in biased research findings. Related studies have shown that incorporating phylogenetic diversity (PD) and functional diversity (FD) provides a complementary and more comprehensive perspective, capable of revealing distinct patterns of biodiversity and their underlying ecological processes (Gianuca et al., 2018). PD refers to the changes in the genetic composition and evolutionary lineage of organisms (Mouquet et al., 2012). FD is defined as the diversity of ecological characteristics shown by organisms within an ecosystem, as well as the ecological functions and services they render (Díaz and Cabido, 2001). FD and PD indicate ecosystem functionality, possible adaptability to environmental alterations, and resilience of biological communities, hence serving as significant supplemental indices for evaluating biodiversity and its correlation with ecosystem processes (Guilhaumon et al., 2015). The patterns exhibited by the same biological communities across different diversity facets (such as phylogenetic and functional) are not always consistent (Zhang et al., 2025). In addition, Strecker et al. (2011) found 75% overlap among diversity priorities, but substantial incongruence remained. This suggests that single-metric approaches fail to capture key facets of biodiversity, particularly in catchments that contribute disproportionately to regional diversity. On account of this, integrating multiple facets diversity indicators for assessment can provide a more comprehensive and reliable understanding of ecological patterns and processes.

As a rule, diversity is assessed by alpha diversity (the number of species at the sampling sites) and beta diversity (variation in species composition among sampling sites). Nonetheless, diversity assessment research has primarily concentrated on alpha diversity (Ge, 2017). In fact, beta diversity is as significant as alpha diversity for ecological theory and applications (Wang et al., 2020), due to its ability to provide complementary insights to those offered by alpha diversity. Beta diversity patterns arise from the replacement of some species by others (turnover) and variations in species richness (nestedness), or the combined effect of both processes (Baselga and Leprieur, 2015; Legendre, 2014). Therefore, analyzing beta diversity helps elucidate relationship between local and regional assemblages, providing crucial insights for understanding diversity patterns across different spatial scales (Socolar et al., 2016).

Key drivers of biodiversity patterns have been extensively explored in numerous studies (Liu et al., 2023; Zhang et al., 2021). Traditionally, research has emphasized the important role of local environmental variables (López-Delgado et al., 2020); however, in recent years, increasing evidence suggests that spatial factors, particularly those related to dispersal limitation and other stochastic processes (e.g., ecological drift and colonization), also have an important impact on variation of aquatic communities (Padial et al., 2014). In general, the role of spatial factors in shaping ecological communities is generally scale-dependent. At large geographic scales, spatial processes tend to be more important than environmental filtering (Zhou et al., 2020). However, the relative importance of these driving factors and their interactions differ among ecosystem types, taxonomic groups, habitats, and spatial scales of research regions (Heino et al., 2015; Soininen et al., 2018). Moreover, environmental and spatial factors exert differing influences on the diversity patterns across different facets (Li et al., 2019). Although previous studies have examined the importance of ecological factors in some freshwater ecosystems (Meng et al., 2025), comprehensive research exploring the relative importance of driving factors of taxonomic, phylogenetic, and functional beta diversity simultaneously remains scarce.

Freshwater fish are essential elements of aquatic ecosystems and hold significant value for the fishery industry and the general population (Zhang et al., 2020). Accurately elucidating the patterns and drivers of freshwater fish diversity is a prerequisite for implementing targeted conservation efforts. As a typical tropical river, the Buyuan River is home to abundant fishery resources and a wide range of fish biodiversity. However, fish diversity in the Buyuan river Basin has suffered from serious devastation under multiple anthropogenic disturbances (e.g., water pollution, river channelization, habitat fragmentation, land-use change, damming, and overfishing) (Li et al., 2024). Compared to historical data, the number of native fish species in the Buyuan River is declining, while the number of invasive fish species is increasing (Fu et al., 2021). While research has investigated fish diversity in the Buyuan River, it has only concentrated on taxonomic diversity and has focused solely on alpha diversity (Fu et al., 2021; Kang et al., 2010). Evidently, previous studies have provided insufficient understanding of the fish diversity patterns in the Buyuan River and their driving factors, which has become a key constraint in formulating effective conservation strategies. To develop effective conservation strategies, it is essential to have a comprehensive understanding of the multifaceted (taxonomic, phylogenetic, and functional diversity) and multilevel (alpha and beta diversity) patterns of fish diversity in the Buyuan River and their driving factors. In addition, the responses of different facets of fish diversity to habitat heterogeneity are inconsistent (Bower and Winemiller, 2019; Pease et al., 2012; Xia et al., 2023). It is therefore necessary to explore the relative importance of the unique habitat characteristics of the Buyuan River in shaping patterns across different diversity facets.

Here, we systematically investigated the diversity patterns of fish assemblages in the Buyuan River Basin, Yunnan, southwestern China. This study aims to investigate the patterns and drivers of multifaceted diversity in fish assemblages of the Buyuan River Basin. Specifically, through this study, we sought to address the following questions:

1) How is the taxonomic, phylogenetic, and functional diversity of fish assemblages in the Buyuan River Basin at the alpha and beta levels?
2) What is the correlation among taxonomic, phylogenetic, and functional diversity?
3) Given that spatial and environmental variables are often highly collinear in a single river ecosystem, it is difficult to fully disentangle their independent effects. Therefore, what is the relative importance of environmental and spatial factors in shaping the multifaceted diversity of the Buyuan River Basin?

2. Materials and methods

2.1. Study area

The Buyuan River Basin (21°40′ - 23°08′ N, 100°52′ - 101°40′ E) is the most important primary tributary of the lower Lancang River and one of the few relatively free flowing tributaries in the basin. It originates from the town of Mohei in Ning'er County, Pu'er City, Yunnan Province, China, and its mainstream is approximately 300 km long, with a catchment area of 8931 km^2 and an elevation range of 500–2290 m in the whole basin (Gu et al., 2019). Influenced by the subtropical monsoon climate, the average annual temperature and precipitation values of the Buyuan River Basin are approximately 21.5 °C and 1488 mm, respectively, and the precipitation is mainly concentrated from June to November (Gu et al., 2019). The good vegetation coverage in the riparian zone of the Buyuan River Basin has resulted in a high diversity of flora and fauna within the basin. Tellingly, the Buyuan River is also a spawning ground and foraging ground for migratory fish in the Lancang-Mekong River Basin (Kang et al., 2010). In the context of hydropower development, the Buyuan River is very important for the maintenance of fish diversity in the Lancang-Mekong River Basin (Bao et al., 2024).

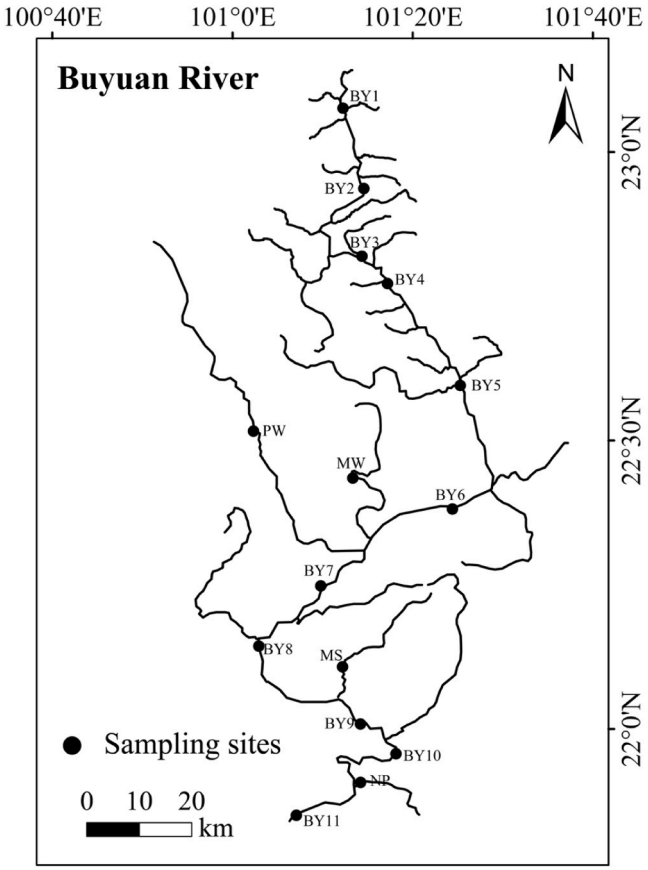


Fig. 1. The sampling sites (black dots) in the Buyuan River Basin in Yunnan Province, China.

2.2. Fish sampling

In April 2023, fish specimens were collected from 15 sampling sites (Fig. 1) across the Buyuan River Basin, including the mainstream and four primary tributaries of the Buyuan River, 11 sites in the Buyuan River (BY1-BY11), 1 site in Mengwang River (MW), 1 site in Puwen River (PW), 1 site in Mansai River (MS), and 1 site in Nanpin River (NP). Sampling sites should be set up in a way that is both habitat accessible and representative, while covering as many habitat types as possible (e.g., riffles, pools). At each sampling site, we collected fish specimens using a backpack electrofishing protocol (12-V import, 250-V export) at the same fishing effort (approximately 45 min for each 200-m river segment). All collected fish specimens were identified to the species level and counted in the field. Voucher specimens were deposited in 10% formaldehyde solution and preserved in the Kunming Institute of Zoology, Chinese Academy of Sciences for further identification in the laboratory. To accurately reflect the native fish diversity of the Buyuan River, non-native species were removed from all subsequent diversity analyses. All species were recorded during the field survey, but the exclusion of non-native taxa ensured that our diversity assessments were not biased by introduced species (Feio et al., 2025). Sampling adequacy was assessed using species accumulation curves, which indicated that coverage reached an asymptote at 15 sites (Figure S1).

2.3. Environmental and spatial factors

In addition to fish sampling, fourteen environmental factors were documented at each sampling site. Water temperature (WT, ℃), pH, dissolved oxygen (DO, mg/L), electrical conductivity (Cond, μS/cm), total dissolved solids (TDS, mg/L), and water nephelometric turbidity units (TU, NTU) were measured in situ using a multiparameter water quality sonde (YSI ProQuatro Multiparameter Meter, YSI Incorporated). River width (RW, m) was measured using a laser rangefinder. Water depth (WD, m) was recorded using a ruler. Flow velocity (FV, m/s), measured with a portable flow meter (Laiende, China), differed significantly across habitat types. Each sampling site's altitude (Alt, m) and distance from the estuary (DFE, km) were obtained from Google Earth software. The slope (Slo) of each sampling site was extracted from the elevation data using ArcGIS software. The percentage of river channelization (PRC) and vegetation coverage (VC) of riparian were assessed using remote sensing satellite data (Li et al., 2024). Among them, the river channelization ratio is a key indicator of the degree of human intervention on rivers, commonly defined as the percentage of the total river length that has been artificially channelized. In remote sensing applications, its calculation essentially involves quantifying morphological metrics such as sinuosity and artificial structures to identify and calculate the proportion of modified river segments relative to the total river length.

Spatial factors were calculated using Moran's eigenvector map (MEM) based on the coordinates (X: Latitude, Y: Longitude). Latitude and longitude were determined using a portable GPS device. MEM was employed to characterize spatial variability across sampling sites by producing spatial variables that represent patterns of overland dispersal from microhabitats (fine scale) to watersheds (broad scale) (Zhou et al., 2020). The obtained eigenvectors were regarded as spatial factors (i.e., Principal Coordinates of Neighbor Matrices (PCNMs)), capable of representing unmeasured broadscale variation in the modern environment or historical influences, such as patterns generated by natural dispersal that illustrate internal local-scale dispersal dynamics or regional-scale migration history (Svenning et al., 2009). PCNMs (a method for extracting multi-scale spatial patterns from site coordinates) generated orthogonal spatial variables from site coordinates, which were used in the model to account for spatial structure and separate environmental from spatial effects on community composition. In this study, only PCNMs (eigenvalues of PCNM components 1–9) possessing positive eigenvalues are utilized as spatial explanatory variables. PCNMs were performed through the pcnm function in the R package vegan.

2.4. Phylogeny and functional traits

Cytochrome c oxidase subunit I (COI) gene sequences of identified fish species were retrieved from the GenBank database on the National Center for Biotechnology Information (NCBI) website (https://www.ncbi.nlm.nih.gov/). The sequences were first aligned using MUSCLE. Molecular phylogenetic trees were subsequently constructed using MEGA11 software based on the neighbor-joining method. Specifically, evolutionary distances were computed using the Kimura 2-parameter model, including both transitions and transversions. To account for rate variation among sites, a gamma-distributed model with a shape parameter of 0.50 was applied, assuming the same pattern among lineages. Node reliability was assessed using the bootstrap method with 10,000 replications. For gaps and missing data, the Partial deletion option was implemented with a 95% site coverage cutoff. Subsequent calculations of phylogenetic diversity were based on the phylogenetic tree constructed here.

For captured fish, we selected seven functional traits (all were categorical variables) to calculate functional diversity. The functional traits selected were mainly related to four aspects of ecological strategies: movement, habitat utilization, feeding, and reproduction (Table S1). All information related to functional traits of fish was obtained from FishBase (https://www.fishbase.org.au/v4). We used fishes of the same genus with similar body shapes and traits to assign species without documented functional traits (Liu et al., 2023). Furthermore, we did not consider any intraspecific variations. For analytical purposes, we presumed that intraspecific variance is significantly less than interspecific variation (McGill et al., 2006).

2.5. Statistical analyses

All analyses were performed utilizing the R software (version 4.5.2, R Core Team, 2022). We first calculated multiple indexes of

alpha and beta diversity using fish abundance data. The alpha facet included taxonomic (species richness (SRic), Pielou, Shannon-Wiener), phylogenetic (Faith's PD, phylogenetic species evenness (PSE), phylogenetic species clustering (PSC)) and functional (functional richness (FRic), functional evenness (FEve), functional divergence (FDiv)) diversity indices. The alpha diversity indices for the three facets were calculated using the vegan, picante, and FD packages in R, respectively. Then we separately calculated the total beta diversity (sor) and divided it into turnover (sim) and nestedness (sne) components across three facets: taxonomic, phylogenetic, and functional. The beta diversity indices for the three facets were calculated as Sørensen dissimilarity based on the incidence data using the betapart package in R.

Spearman's rank correlations were calculated using the cor function from the R package corrplot to explore the relationships among taxonomic, phylogenetic, and functional alpha diversity. A Mantel test was employed to investigate the congruence among taxonomic, phylogenetic, and functional beta diversity and their turnover and nestedness components. The Mantel tests were performed using the mantel function of the vegan package with 9999 permutations in R.

To explore the drivers of alpha diversity while avoiding overfitting, environmental variables (except pH) underwent Z-score standardization and were subsequently analyzed by principal component analysis (PCA). The first several principal components (denoted as Env_PC) with a cumulative variance explained of $\geq$ 70% were retained as environmental predictors. Concurrently, spatial variables were screened using forward stepwise regression combined with the akaike information criterion (AIC). The final predictor matrix (Table S3) comprised the environmental PC axes and the selected PCNM variables. Spearman's rank correlation analysis was

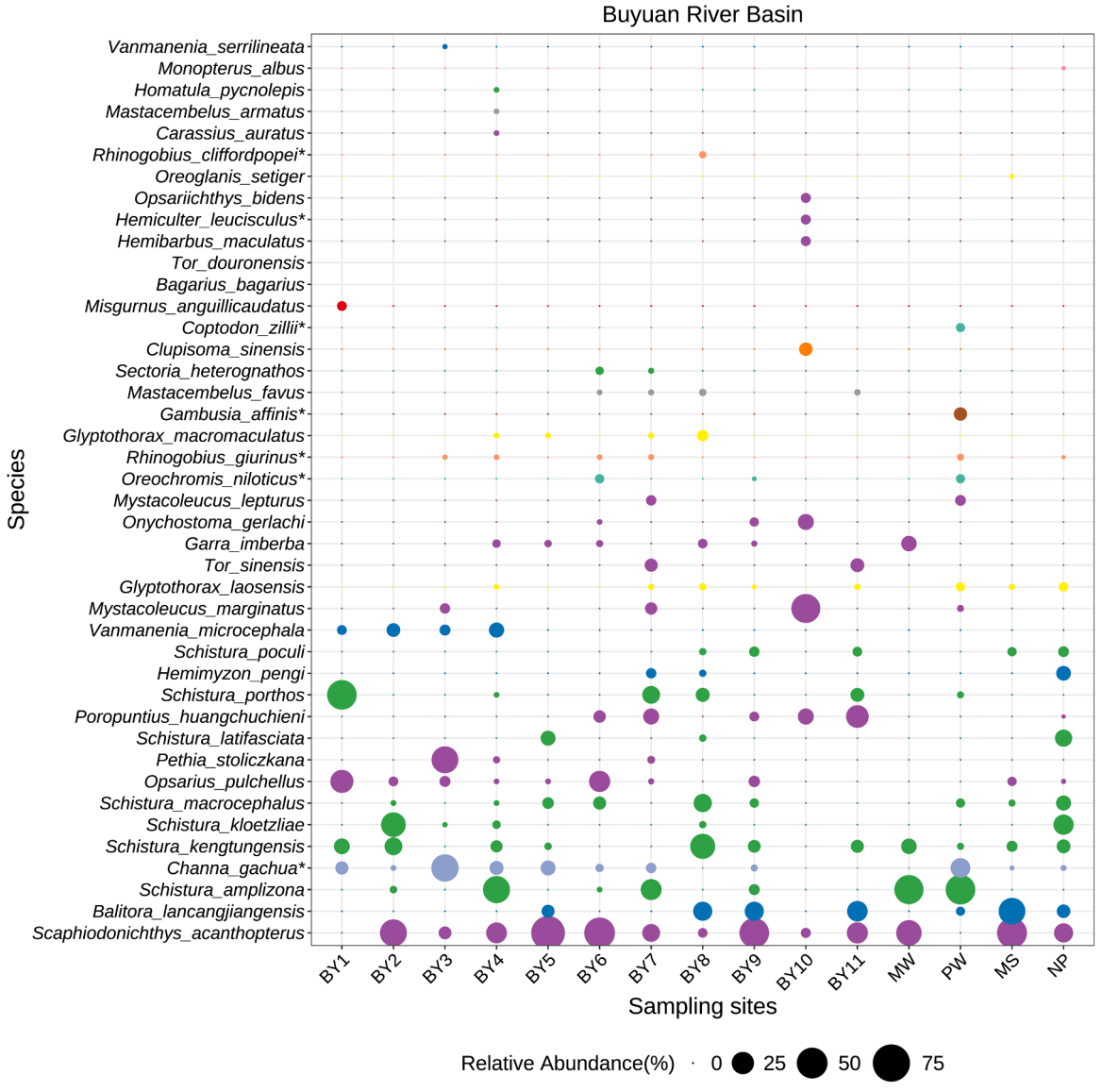


Fig. 2. Bubble chart of the species composition of fish communities at different sample sites in the Buyuan River Basin at the species level. Colors denote the family affiliation of the fish. Fish marked with an asterisk (*) are non-native species.

employed to examine the monotonic relationships between each alpha diversity index and each predictor, yielding Spearman's rho and uncorrected P values. The Benjamini-Hochberg (BH) method was applied for multiple testing correction to obtain FDR-corrected P values. A threshold of FDR < 0.05 was considered statistically significant, and all tests were two-tailed. The Spearman rank correlation was performed using the cor.test function from the stats package in R, with the argument method = "spearman". The FDR correction was performed using the p.adjust function from the stats package, with the argument method = "BH".

To mitigate the risk of overfitting, the same screened explanatory variable matrix as that used in the alpha diversity analysis was employed to investigate the drivers of beta diversity. Distance-based redundancy analysis (db-RDA) combined with a variance partitioning analysis (VPA) were employed to elucidate the relative importance of environmental and spatial factors on taxonomic, phylogenetic and functional beta diversity and their components (Peres-Neto et al., 2006). dbRDA was performed using the capscale function of the vegan package, and VPA was conducted using the function varpart in vegan. In addition, the significance of pure fractions of predictor variables was tested using the function anova.cca in vegan with 999 permutations.

3. Results

3.1. Species composition

A total of 1593 fish specimens comprising 42 species (Fig. 2) were collected in the Buyuan River Basin, which belong to 5 orders, 12 families, and 29 genera (Table S2). Within the 5 orders, Cypriniformes was the most dominant, covering 66.7% of the fish species, followed by Siluriformes (11.9%) and Perciformes (11.9%) (Figure S2). At the species level, the species with high relative abundance were *Scaphiodonichthys acanthopterus*, *Balitora lancangjiangensis*, *Schistura amplizona*, and *Channa gachua* (Fig. 2). By contrast, *Vanmanenia serrilineata*, *Monopterus albus*, *Homatula pycnolepis*, and *Mastacembelus armatus* were found to have low relative abundance and were detected at low frequency across the sampling sites (Fig. 2).

3.2. Patterns of multi-faceted fish diversity

The multi-faceted alpha diversity indices of fish in the Buyuan River are shown in Table 1. For taxonomic diversity, the average species richness index was 9 ± 3, the average Pielou and Shannon-Wiener indices were 0.73 ± 0.11 and 1.57 ± 0.37, respectively. For phylogenetic diversity, the average Faith's PD index was 1.80 ± 0.66, the average PSE and PSC indices were 0.64 ± 0.10 and 0.53 ± 0.13, respectively. For functional diversity, the average FRic index was 10.73 ± 7.83, the average FEve and FDiv indices were 0.51 ± 0.12 and 0.8 ± 0.09, respectively.

Patterns of beta diversity differed significantly across multiple facets (Fig. 3). The mean values of taxonomic (βsor.tax), phylogenetic (βsor.phy), and functional (βsor.fun) total beta diversity of fish communities in the Buyuan River were 0.61 ± 0.09, 0.44 ± 0.15, and 0.83 ± 0.08, respectively (Fig. 3). For taxonomic and phylogenetic beta diversity, the difference in total beta diversity was mainly caused by their turnover component (βsim.tax, mean value: 0.50 ± 0.12, accounting for 82.0%; βsim.phy, mean value: 0.31 ± 0.13, accounting for 70.5%; Fig. 3). For functional beta diversity, the differences in the total beta diversity were mainly driven by its nestedness component (βsne.fun, mean value: 0.44 ± 0.17, accounting for 53.0%; Fig. 3).

3.3. Correlations among multi-faceted fish diversity

At the alpha diversity level, the pairwise comparisons of each biodiversity index revealed varying correlations among the indices (Fig. 4). Specifically, species richness was positively correlated with Shannon-Wiener ($r = 0.81$, $p < 0.001$) and FRic ($r = 0.79$,

Table 1
Multi-faceted alpha diversity index of fish at each sampling site in the Buyuan River Basin.

Sites	SRic	Pielou	Shannon-Wiener	Faith's PD	PSE	PSC	FRic	FEve	FDiv
BY1	5	0.76	1.23	1.87	0.55	0.58	5.71	0.57	0.95
BY2	7	0.76	1.47	1.98	0.71	0.55	4.56	0.38	0.70
BY3	7	0.65	1.26	1.99	0.67	0.81	5.49	0.66	0.82
BY4	15	0.66	1.79	2.58	0.65	0.52	30.52	0.51	0.73
BY5	8	0.55	1.15	2.02	0.52	0.63	6.84	0.41	0.88
BY6	9	0.62	1.36	0.68	0.59	0.50	11.29	0.60	0.63
BY7	14	0.83	2.20	2.30	0.72	0.50	19.81	0.54	0.79
BY8	13	0.79	2.02	2.17	0.63	0.30	8.51	0.45	0.82
BY9	11	0.72	1.73	2.17	0.70	0.57	9.66	0.52	0.76
BY10	9	0.79	1.75	0.76	0.62	0.64	16.90	0.73	0.85
BY11	9	0.85	1.86	2.06	0.79	0.60	16.82	0.69	0.76
MW	4	0.88	1.21	0.29	0.66	0.47	0.17	0.39	0.81
PW	8	0.55	1.15	1.82	0.37	0.41	3.66	0.50	0.85
MS	8	0.60	1.24	2.05	0.68	0.44	6.24	0.41	0.96
NP	12	0.86	2.13	2.25	0.73	0.36	14.80	0.32	0.74

SRic: species richness, PSE: phylogenetic species evenness, PSC: phylogenetic species clustering, FRic: functional richness, FEve: functional evenness, FDiv: functional divergence, BY: Buyuan River, MW: Mengwang River, PW: Puwen River, MS: Mansai River, NP: Nanpin River.

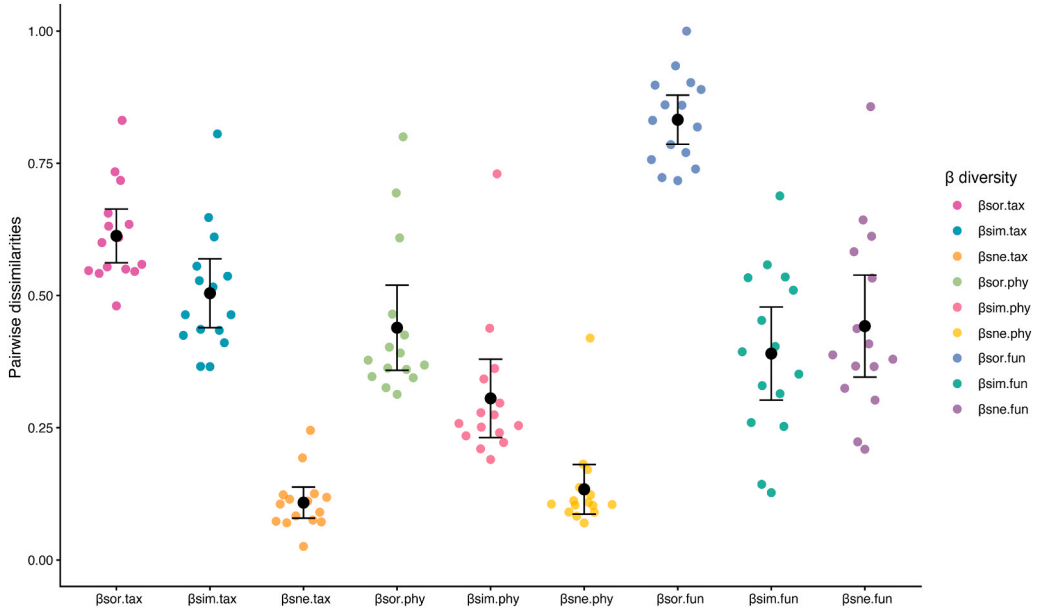


Fig. 3. Beeswarm plot of the different components of taxonomic, phylogenetic, and functional beta diversity of fishes in the Buyuan River Basin. Black dots represent the mean value of each component, and the deviation lines represent the 95% confidence intervals of the group means. βsor represent the total beta diversity, βsim represent the turnover component of beta diversity, βsne represent the nestedness component of beta diversity. tax is shorthand for taxonomic, phy is shorthand for phylogenetic, fun is shorthand for functional.

$p < 0.001$) (Fig. 4). Moreover, Shannon-Wiener, a composite index that integrates both richness and evenness, showed a positive correlation with the FRic ($r = 0.67$, $p < 0.01$) (Fig. 4). In addition to these, the correlation between other biodiversity indices was relatively weak (Fig. 4).

At the beta diversity level, the results of the Mantel tests displayed a significant correlation between taxonomic, phylogenetic, and functional diversity (Fig. 5). The strongest relationships were found between taxonomic diversity and phylogenetic diversity, with values of correlation coefficients ranging from 0.65 to 0.88 (Fig. 5a-c). Further, we found relatively weaker correlations between taxonomic diversity and functional diversity (Mantel's r from 0.40 to 0.54) (Fig. 5d-f), as well as between phylogenetic diversity and functional diversity (Mantel's r from 0.36 to 0.52) (Fig. 5g-i).

3.4. Ecological drivers of multi-faceted fish diversity

The Spearman correlation heatmap indicated that after BH-FDR correction, none of the correlations between multifaceted alpha diversity indices and explanatory variables reached statistical significance (FDR > 0.05), with only trend-level associations observed (Fig. 6). Specifically, Env_PC1 showed relatively strong positive trends with Shannon-Wiener (rho = 0.56, P = 0.031) and SRic (rho = 0.52, P = 0.047). Env_PC4 showed positive trends with PSC, Shannon-Wiener, and FRic. PCNM1 showed positive trends with most alpha diversity indices. PCNM4 exhibited a relatively strong negative correlation with FRic (rho = −0.76, P = 0.001), and PCNM6 was negatively correlated with SRic, PD, and Shannon-Wiener. Overall, Env_PC1, Env_PC4, and PCNM1 were positively correlated with multifaceted alpha diversity indices, whereas PCNM4 and PCNM6 were negatively correlated with them.

The results of the VPA indicate that the explanatory variables exert differing effects on beta diversity across three facets (Fig. 7). For overall beta diversity, the explanatory variables explained 28.3% and 29.8% of the variation in taxonomic and functional diversity, respectively. For the turnover component of beta diversity, the interpretation rates of the explanatory variables were 26.6% for taxonomic diversity, 12.4% for functional diversity, and 5.6% for phylogenetic diversity. For the nestedness component of beta diversity, the explanatory variables explained 8.1% and 5.1% of the variation in taxonomic and functional diversity, respectively. For taxonomic and functional diversity, the pure environmental variables explained more of the variation in total beta diversity and its turnover component than did the spatial variables. However, the environmental variables explained less of the variation in the nestedness component than did the spatial variables. For phylogenetic diversity, the explanatory variables explained none of the variation in total beta diversity or its nestedness component. In addition, although pure environmental variables explained more of the variation in the turnover component than did spatial variables, the overall proportion of variation explained was very low (5.6%).

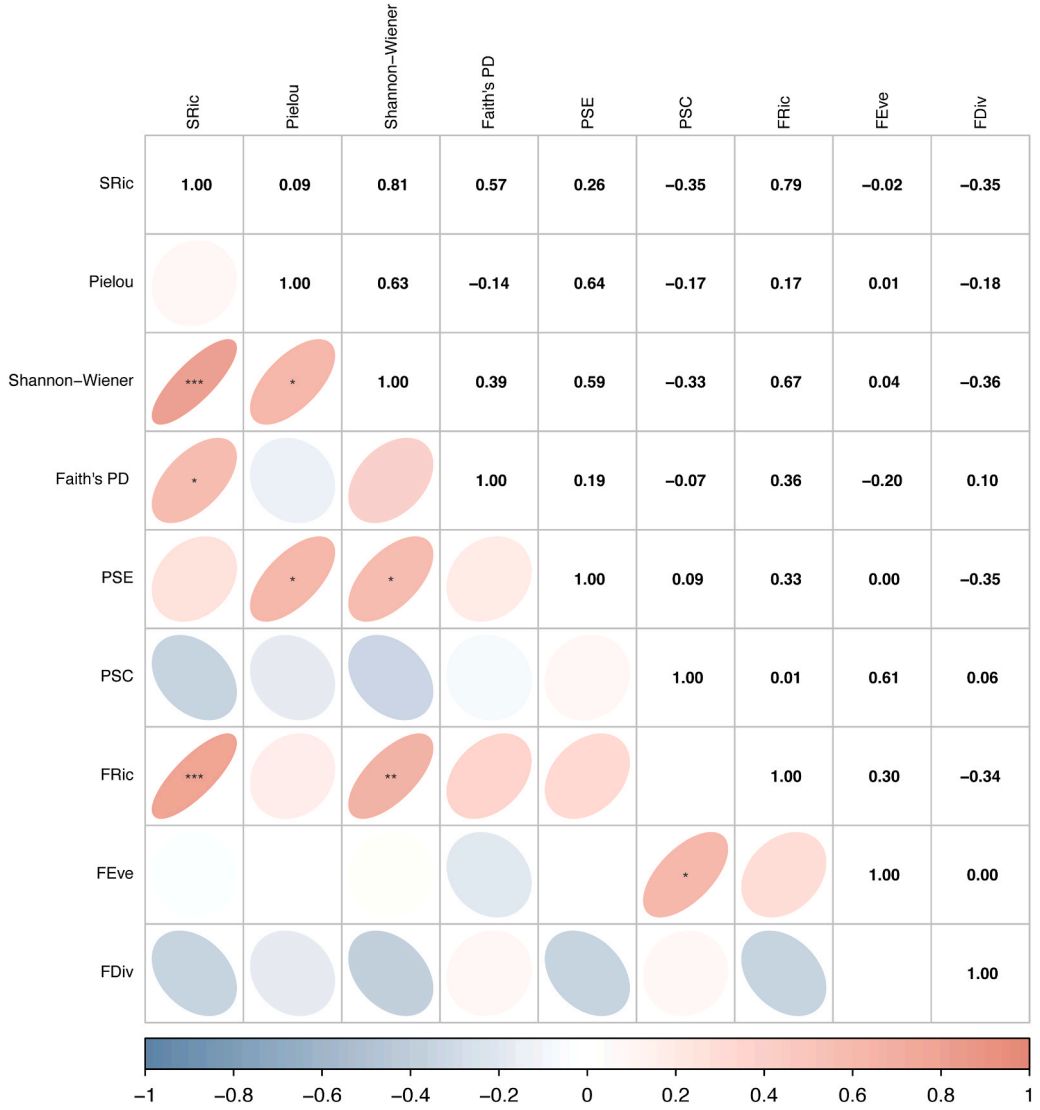


Fig. 4. Correlations among multi-faceted fish diversity at the alpha levels. The upper triangle of the plot presents the correlation coefficients for each index pair, whilst the lower triangle displays the associated significance (p value). ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$. SRic, species richness; PSE: phylogenetic species evenness, PSC: phylogenetic species clustering, FRic: functional richness, FEve: functional evenness, FDiv: functional divergence.

4. Discussion

4.1. Fish diversity patterns

By revealing the multiple facets diversity patterns of fish assemblages in the Buyuan River Basin, this study lays a scientific foundation for the conservation of fish diversity in the basin. Analysis of alpha diversity across three facets revealed significant differences in index values among sampling sites (Table 1), highlighting the complexity and high heterogeneity of fish distribution patterns in the Buyuan River Basin. This may be related to the complex and diverse habitats within the Buyuan River Basin, as habitat heterogeneity and energy availability are key factors determining species richness (Guégan et al., 1998). Although the Buyuan River Basin has suffered significant human disturbance, some reaches of the river remain in relatively pristine condition due to their remote geographical location. An investigation into the relationships among alpha diversity indices across three facets revealed a significant positive correlation between species richness and FRic index (Fig. 4). This indicates that as species richness increases, niche space also becomes more fully occupied (Mason et al., 2005). The significant positive correlation between the Shannon-Wiener index and the FRic index (Fig. 4) is still influenced by species richness. Given that high Shannon-Wiener index values indicate greater species richness within a community, and FRic is often constrained by species richness (Mason et al., 2005; Villéger et al., 2008). Beyond this, the

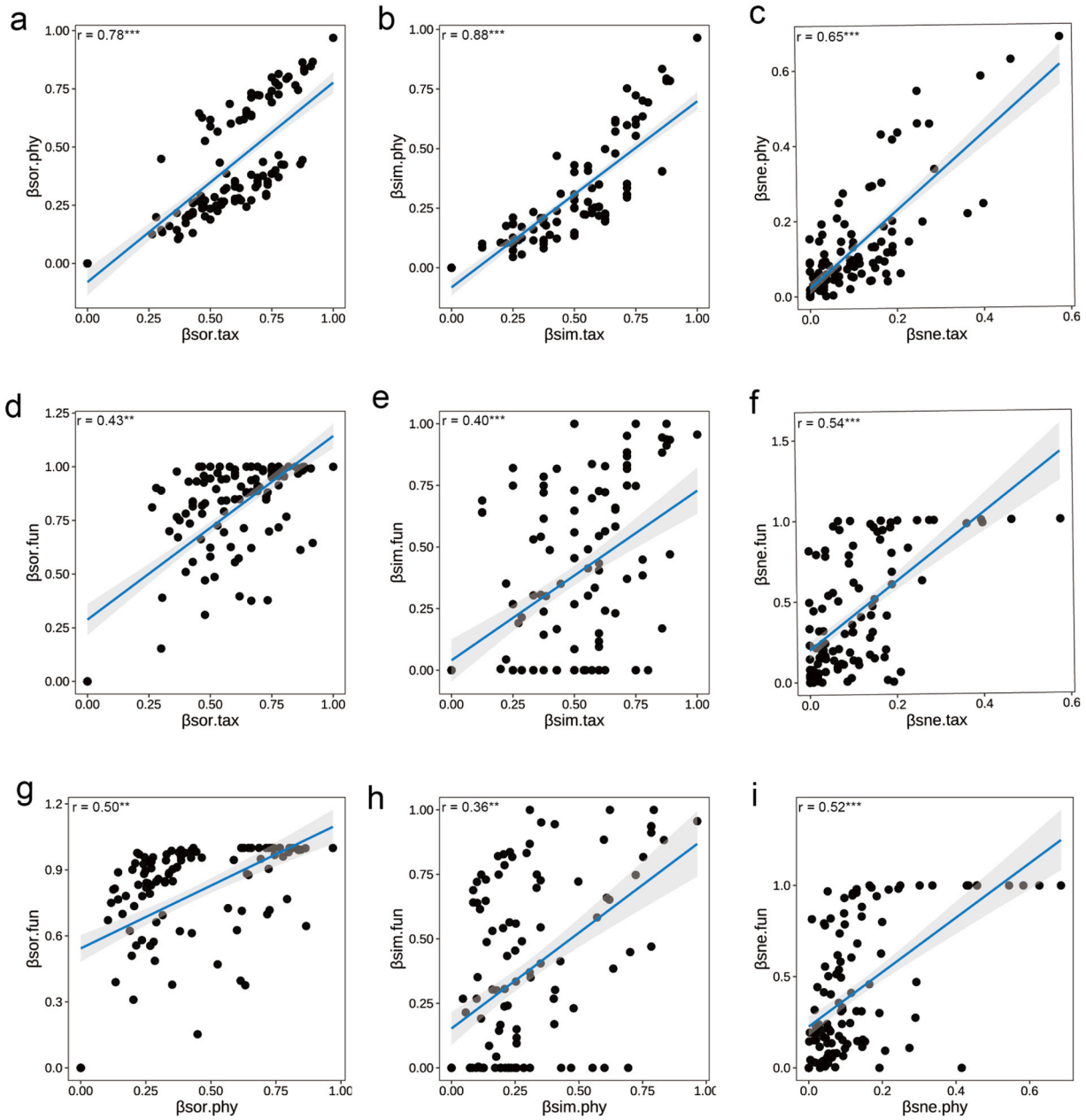


Fig. 5. Relationships among multi-faceted fish beta diversity components. Correlation (Mantel's r) and significance were shown for each relationship. βsor represent the total beta diversity, βsim represent the turnover component of beta diversity, βsne represent the nestedness component of beta diversity. tax is shorthand for taxonomic, phy is shorthand for phylogenetic, fun is shorthand for functional. ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$.

correlations among other alpha diversity indices are relatively weak. This finding confirms the perspective that ecological information provided by diversity indices across different facets is complementary and has distinct focuses (Corbelli et al., 2015; Heino and Tolonen, 2017b). That is, single-facet diversity index struggles to accurately predict or substitute for diversity patterns across other facets. This provides us with an important insight: when evaluating biodiversity, we should comprehensively consider diversity across multiple facets to fully reflect the current situation of biodiversity within the study area.

Beta diversity evaluates the variation in species composition between communities, as opposed to alpha diversity, which measures species richness within a single community (Legendre and De Cáceres, 2013). Through our research, we found that the beta diversity of taxonomic and phylogenetic facets in the Buyuan River Basin is driven by its turnover component, while functional beta diversity is driven by its nestedness component (Fig. 3). The findings are consistent with certain earlier research (Liu et al., 2023; Perez Rocha et al., 2018; Wu et al., 2021). Selective extinction may be one of the major causes of functional beta diversity being driven by its

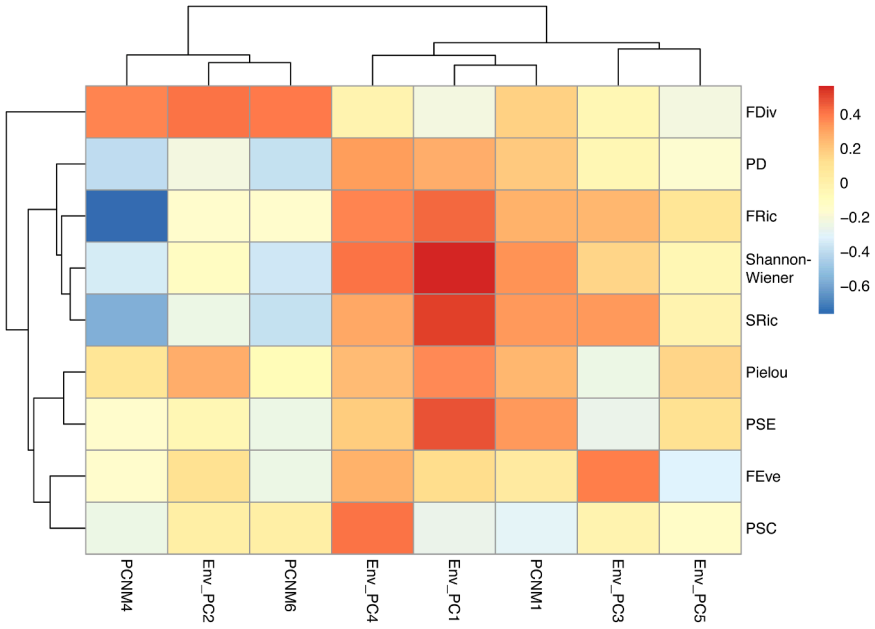


Fig. 6. Heatmap of Spearman correlations between alpha diversity indices and explanatory variables. Colors represent Spearman's rho (red, positive correlation; blue, negative correlation). SRic, species richness; PSE: phylogenetic species evenness, PSC: phylogenetic species clustering, FRic: functional richness, FEve: functional evenness, FDiv: functional divergence.

nestedness component, as environmental filtering typically makes certain traits more common than others (Wu et al., 2021). However, this reason is only an inference based on previous studies, and the single survey in this study cannot verify it. In addition, variations in habitat heterogeneity among sampling sites within the watershed also contribute to functional beta diversity being driven by nestedness (Liu et al., 2023). Sampling sites with high habitat heterogeneity harbor species exhibiting diverse traits, whereas species within sites characterized by severe habitat homogenization tend to display certain preferences in their traits (Goldenberg Vilar et al., 2014). Therefore, the functional traits of species within homogeneous sites are typically a subset of the functional traits of species within highly heterogeneous sites (Heino and Tolonen, 2017a). Taxonomic and phylogenetic beta diversity are driven by turnover components, indicating that some sampling sites within the study area exhibit differences in species composition (Hill et al., 2019).

Exploring the relationships among taxonomic, functional, and phylogenetic diversity at both watershed and local scales facilitates a deeper understanding of the ecological connections among the various components of beta diversity (Huang et al., 2020; Steudel et al., 2016). The results of the Mantel tests revealed an extremely strong correlation between taxonomic and phylogenetic beta diversity, indicating that closely related species exhibit greater morphological similarity than distantly related species (Xia et al., 2023). This is consistent with findings from previous studies (Qian et al., 2021; Tsianou et al., 2021; Wu et al., 2022; Xia et al., 2023). This may be related to the fact that both taxonomic and phylogenetic beta diversity are driven by their turnover components (Xia et al., 2023). Moreover, spatial scales and human disturbance may also lead to a strong correlation between taxonomic and phylogenetic beta diversity (Wu et al., 2022). The weaker correlation between taxonomic and functional beta diversity, as well as between phylogenetic and functional beta diversity, stems from the complementarity of the information reflected by diversity across different facets (Branco et al., 2020; HEINO, 2011). Our findings underscore the necessity of integrating multiple facets in biodiversity assessments, as the ecological information provided by these facets is complementary.

4.2. *Drivers of fish diversity patterns*

Understanding the drivers of biodiversity patterns is a necessary prerequisite for effective biodiversity conservation. Relevant studies indicate that fish community structure is shaped by the combined effects of environmental and spatial factors, but their relative importance varies among different facets of diversity (Li et al., 2019). We found that environmental and spatial variables influenced the alpha diversity of fish in the Buyuan River across three facets to some extent (Fig. 6). For example, relevant studies indicate that both channelization of rivers and changes in riparian vegetation cover can influence the species composition and diversity of fish communities (Bonk and Bobrek, 2021; Torres-Bejarano et al., 2022). Indicators reflecting habitat heterogeneity, such as river width, water depth, and flow velocity, are also closely related to fish diversity (Gorman and Karr, 1978; Vieira and Tejerina-Garro, 2020). Similarly, water quality also has a crucial impact on fish diversity. Decreased dissolved oxygen concentration inhibits metabolism and can bring sublethal effects to organisms in severe cases (Jacobsen, 2008), thereby affecting the species composition within the community. Increased water turbidity can cause visual disturbance to fish. Water temperature affects fish metabolism, growth, reproduction, and distribution (Magnuson et al., 2015). Altitude influences the patterns of fish diversity by regulating key water

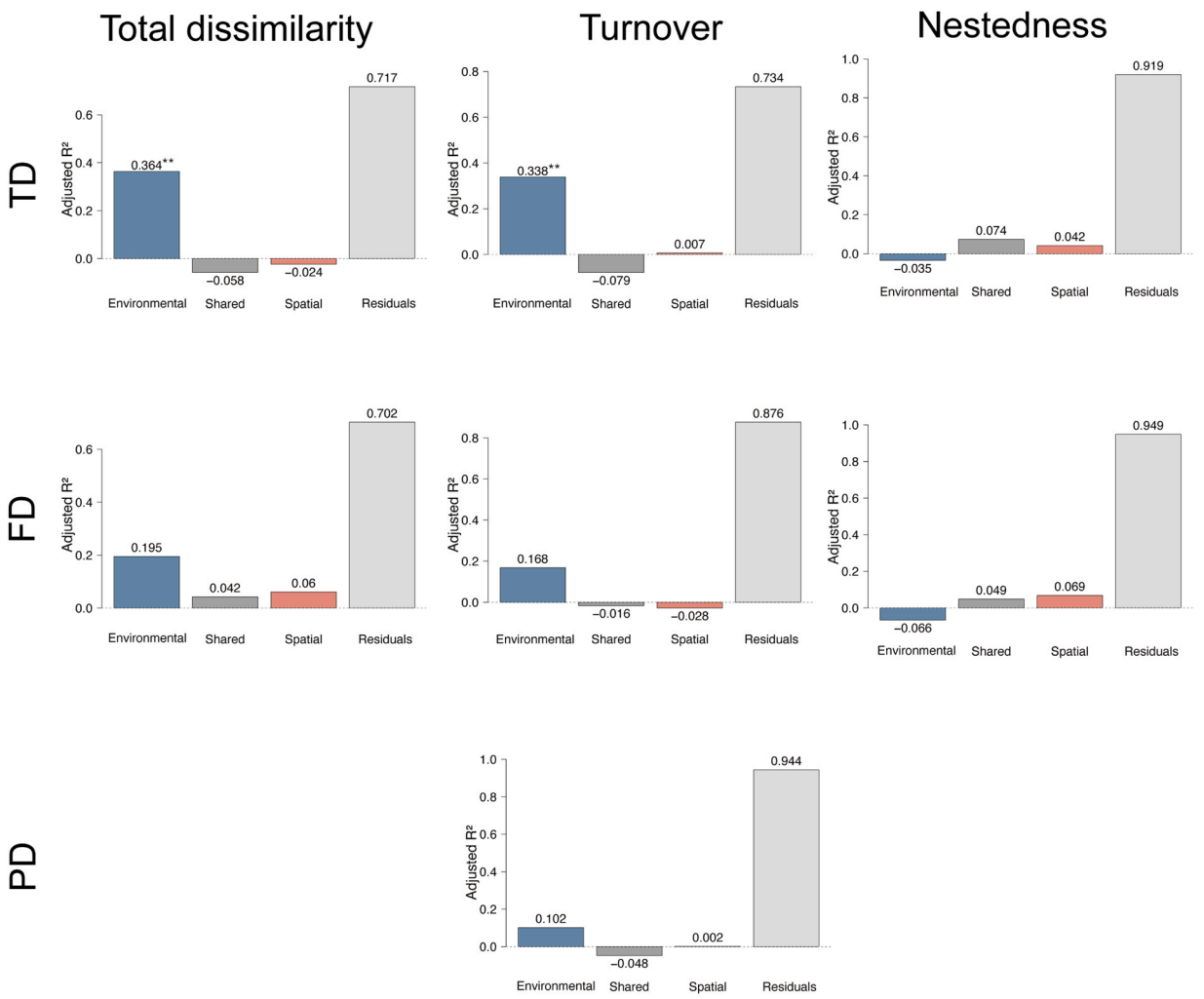


Fig. 7. Results of variation partitioning analysis showing the effects of explanatory variable groups on the taxonomic, phylogenetic, and functional beta diversity of fish and their components (turnover and nestedness) in the Buyuan River Basin. TD: taxonomic diversity, PD: phylogenetic diversity, FD: functional diversity. ***: $p < 0.001$, **: $p < 0.01$, *: $p < 0.05$.

environment parameters, including water temperature, dissolved oxygen levels and flow velocity (Fang et al., 2021). According to the river continuum concept, the riverine environment exhibits a continuous gradient from headwater to estuary (Vannote et al., 1980). Therefore, the distance from the estuary of sampling sites is a key factor determining fish diversity patterns. In addition, spatial variables often exert some influence on fish taxonomic and phylogenetic diversity due to dispersal limitation and historical processes (Li et al., 2019). As geographical conditions change along the longitudinal river gradient, fish functional trait composition shifts accordingly, and spatial variables therefore also have some effect on fish functional diversity (Schmera et al., 2013). In summary, the three facets alpha diversity of fish in the Buyuan River is shaped by the combined effects of all explanatory variables.

The results of dbRDA and VPA indicated that the fish taxonomic and functional beta diversity in the Buyuan River Basin was jointly driven by environmental and spatial variables (Fig. 7). It should be noted that, for taxonomic and functional beta diversity, as well as for the turnover components of both, environmental factors are more important than spatial factors. This is in accordance with the outcomes of previous investigations (Zhang et al., 2021). The relative importance of environmental and spatial factors in shaping fish diversity patterns is scale dependent (da Silva et al., 2026; Loyola-Bartra et al., 2023). Typically, environmental gradients are more pronounced at larger spatial scales (Xia et al., 2023). In the Buyuan River Basin, habitat conditions for fish change distinctly from upstream to downstream, filtering species adapted to different environmental conditions. This largely explains why environmental variables are relatively more important than spatial variables in driving the spatial heterogeneity of fish species composition. In addation, different species may exhibit similar traits across broad temporal or spatial scales (Villéger et al., 2013), and traits related to environmental tolerance and life history can influence the possibility of some species appearing in a community (Arantes et al., 2018). Thus, environmental factors exert a stronger influence on functional beta diversity than spatial factors. Additionally, the fact that phylogenetic beta diversity was insensitive to all explanatory variables in this study may be due to weak phylogenetic signal, or may be

an artifact arising from statistical methods (e.g., collinearity and unmeasured variables) or scale effects. Further studies are needed to confirm this phenomenon. Finally, it is worth noting that VPA cannot fully explain the variation in the fish community structure of the Buyuan River. This may be related to unmeasured variables (e.g., river connectivity, hydrological regime, fishing pressure, unobserved biological effects, etc.) (Liu et al., 2023; Zhou et al., 2020).

4.3. Implications for biodiversity conservation

In the Buyuan River Basin, taxonomic and phylogenetic β-diversity were primarily driven by turnover, whereas functional beta diversity was mainly determined by nestedness. These findings suggest that conservation strategies for fish diversity in this region should focus on protecting multiple sites to capture the unique species composition contributed by turnover, while also prioritizing sites with high functional alpha diversity to preserve functionally important assemblages. Specifically, site BY8 had the third highest species richness and phylogenetic diversity among all sites, whereas its functional richness was relatively low (Table 1). In contrast, site BY10, which had the third highest functional richness, exhibited low phylogenetic diversity (Table 1). From the perspective of fish diversity conservation, both BY8 and BY10 are of high conservation value. This phenomenon arises precisely from the relative independence among different facets of diversity. In addition, the influences of environmental and spatial factors on beta diversity varied across the three diversity facets. Accordingly, effective conservation of fish diversity in this region requires both habitat heterogeneity and longitudinal connectivity. Given that the Buyuan River serves as a spawning ground for numerous migratory fish species in the Lancang-Mekong River Basin, it is imperative to maintain the longitudinal connectivity of this basin to ensure unobstructed migratory corridors for fish. In addition, human alterations to the river landscape should be minimized as much as possible. This would help maintain relatively pristine habitats and support fish species with diverse habitat preferences. Overall, our findings emphasize the importance of considering multiple facets of diversity in the development of biodiversity conservation strategies.

4.4. Limitations of the study

Riverine fish communities display temporal dynamics, with variations occurring at seasonal and interannual scales. It must be acknowledged that a single-season survey can only provide a snapshot at a specific point in time, and is therefore prone to missing seasonally occurring species. In this study, fish sampling was conducted solely during the dry season, which might not detect migratory fish species that appear only in the wet season. This may lead to an underestimation of species richness, functional richness, and phylogenetic diversity within the study region. In addition, the rise in water levels during the normal-flow and wet seasons enhances river connectivity. This increased connectivity not only provides fish with more food resources and habitat conditions but also facilitates their dispersal along the river network (Jiang et al., 2020), thereby weakening the relative importance of environmental filtering in shaping fish beta diversity during the high-water period. Correspondingly, the patterns of fish beta diversity would also change. Enhanced dispersal increases the similarity among local fish communities (de Macedo-Soares et al., 2010) and simultaneously reduces the spatial heterogeneity of phylogenetic diversity (Xia et al., 2023). However, rising water levels may also reshape environmental gradients, thereby exerting differential filtering effects on fish with different functional traits and consequently increasing the spatial heterogeneity of functional diversity (Cabana et al., 2017). Therefore, relying solely on dry-season survey data may bias estimates of fish beta diversity in the Buyuan River Basin, leading to an overestimation of taxonomic and phylogenetic beta diversity and an underestimation of functional beta diversity. Further empirical studies across multiple seasons are needed to verify these patterns. Although this factor may affect the precision of the fish diversity assessment, one of the core objectives of this study is to investigate the interrelationships among multiple dimensions of diversity. In this regard, this scientific question can still be adequately resolved. In general, for the evaluation of fish diversity, it is necessary to conduct multiple surveys to monitor the full range of species present in the study area as comprehensively as possible, thereby yielding the most realistic outcomes.

5. Conclusions

Investigating the patterns of diversity across multiple facets and their driving factors not only uncovers spatial patterning of ecological communities but also facilitates the prediction of potential alterations in community structure and ecosystem functions (Zhang et al., 2021). Here, we investigated the patterns and driving factors of taxonomic, phylogenetic, and functional diversity in the Buyuan River Basin. The results reveal that different facets of diversity exhibit distinct patterns and driving factors. Additionally, diversity across different facets exhibits a degree of independence, and the information they provide is complementary. In short, the information provided by diversity in any one facet cannot be substituted by diversity in other facets. Our research underscores the importance of employing three-facet diversity indicators when conducting biodiversity assessments.

CRediT authorship contribution statement

Miao Li: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jiang Bai:** Visualization, Methodology, Formal analysis, Data curation. **Chenyang Yao:** Methodology, Investigation, Data curation. **Xuanyu Wang:** Methodology, Data curation. **Xiaopeng Cheng:** Visualization, Methodology, Formal analysis. **Guiquan Fu:** Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Deping Kong:** Validation, Supervision, Project administration, Methodology, Conceptualization. **Xiaoyong Chen:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Ethics approval statement

All methods used in this study were conducted in accordance with the Laboratory Animal Management Principles of China. All experimental protocols in this study were approved by the Ethics Committee for Animal Experiments of the Kunming Institute of Zoology, Chinese Academy of Sciences.

Declaration of Competing Interest

The authors assert that they possess no identifiable personal affiliations or conflicting financial interests that would have exerted an influence on the research undertaken in this study.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.gecco.2026.e04477.

Data availability

Data will be made available on request.

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